AWS Lambda offers a maintenance-free and highly available computing environment. You no longer need to install security updates, replace failed virtual machines, or manage remote access (such as SSH or RDP) for administrators. On top of that, AWS Lambda is billed by invocation. Therefore, you don't have to pay for idling resources that are waiting for work (e.g., for a task triggered once a day).

In our first example, you will create a Lambda function that performs periodic health checks for your website. This will teach you how to use the Management Console to get started with AWS Lambda quickly. In our second example, you will learn how to write your own Python code and deploy a Lambda function in an automated way using CloudFormation, which we introduced in chapter 4. Your Lambda function will automatically add a tag to newly launched EC2 instances. At the end of the chapter, we'll show you additional use cases like building web applications and Internet of Things (IoT) backends and processing data with AWS Lambda.

Examples are almost all covered by the Free Tier

The examples in this chapter are mostly covered by the Free Tier. There is one exception: AWS CloudTrail. In case you already created a trail in your account, additional charges—most likely just a few cents—will apply. For details, please see https://aws.amazon.com/cloudtrail/pricing/.

You will find instructions on how to clean up the examples at the end of each section.